THOMAS BERNARD



French

Trento

Driving license

thomas.bernard.consulting@gmail.com

TECHNICAL SKILLS

- Fluvial morphodynamics| landslides
- Natural-hazards analysis | topographic change detection
- Hydraulic modelling 1D/2D | statistical modelling
- DTM/DSM creation
- 3D data processing (LiDAR topo-bathimetric, photogrammetry)

SOFTWARES & TOOLS

- Python (advanced) | Jupyter Notebook | Github
- Rasterio | Geopandas | GDAL
- Machine & Deep learning
- HEC-RAS | Graphflood
- QGIS | CloudCompare | Lastools
- Microsoft suite

SOFT SKILLS

- Project & budget management
- International communication
- Autonomy, adaptability, fast-learner, responsible

LANGUAGES

French | Mothertongue English | C2
Italian | C1 German | A2

Environmental And Geospatial Engineer

Expert in Natural-hazards, Hydromorphology and Geospatial analyses

I am an Environmental researcher and computational specialist with a PhD in Environmental Sciences and a strong foundation in natural-hazards and geospatial analysis. I am using advanced Python, modelling, GIS and both 2D/3D data processing to understand natural processes and assess related risks. Backed by extensive experience in processing and analyzing geospatial data, recently enhanced by specialized training in machine and deep learning applied to 2D and 3D datasets, I am eager to apply my technical expertise to advance innovative solutions for environmental monitoring.

PROFESSIONAL EXPERIENCES

Jun 2026 - Jul 2026

Hydraulic engineer (consultancy)

Vesta Energy srl – Trento, Italy

- Simulation of 100-year floods to **assess the impact of infrastructure in flood-prone areas**
- Technical standards of the PGRA 2021-2027
- Responsible of the Hydraulic Compatibility Assessment Study

Oct 2023 - Apr 2026

Post-doc researcher in Hydromorphology

University of Trento - Italy

- Management of an applied research project in river morphodynamics
- **1D/2D hydraulic and sediment transport modelling** (numerical and physical)
- Preparation of technical reports and coordination with international partners
- Organization of a technical workshop on the management of Alpine rivers (planning, funding, coordination)

Apr 2022 - Jan 2023

Remote sensing & Geospatial engineer

University of Rennes - France

- Processing and classification of **topo-bathymetric LiDAR** data using **machine learning**
- Reconstruction of river bathymetry using **2D hydraulic simulations**
- Production of high-resolution digital terrain models (DTMs)

Oct 2018 - Mar 2022

Junior researcher in Natural-hazards and Remote sensing

University of Rennes - France

- 2D hydraulic simulations for **floodplain mapping and risk analysis**
- Development of workflows for detecting **topographic changes** and landslides
- LiDAR data processing and classification
- Production of technical reports | international communication
- Teaching activity >100 hours

EDUCATION

May 2026 - Today

3D AI Architect
Machine & deep learning applied to 3D datasets
3D Geodata Academy
Florent Poux

Mar 2022

PhD in Environmental sciences
University of Rennes - France

July 2018

Master's Degree in Earth sciences
University of Rennes - France

SCIENTIFIC PUBLICATIONS

- **2026 (in review):** Nativ, R., Lague, D., Leroy, P., Guerit, L., **Bernard, T.,** Letard, M., ... & Steer, P. (2026). From point cloud to digital elevation model: Airborne topo-bathymetric LiDAR processing over the 28 km Ardèche River Gorges. *EGUsphere*, *2026*, 1-47.

- **2026 : Bernard, T. G**., Pandrin, E., Capito, L., Bizzi, S., & Bertoldi, W.: "*Quantification of the active width in gravel-bed rivers: the effect of morphology, flood magnitude and survey frequency*". *Geomorphology.*

- **2024 :** Gailleton, B., Steer, P., Davy, P., Schwanghart, W., & **Bernard, T.** (2024). GraphFlood 1.0: an efficient algorithm to approximate 2D hydrodynamics for landscape evolution models. *Earth Surface Dynamics*, *12*(6), 1295-1313.

- **2022 : Bernard, T. G.,** Davy, P., & Lague, D. : "*Hydro-geomorphic metrics for high resolution fluvial landscape analysis*". *Journal of Geophysical Research: Earth Surface*, *127*(3), e2021JF006535. (**Top cited article** of 2022-2023)

- **2021 : Bernard, T. G.,** Lague, D., & Steer, P. : "*Beyond 2D landslide inventories and their rollover: synoptic 3D inventories and volume from repeat lidar data*". Earth Surface Dynamics, 9(4), 1013-1044.

PhD thesis: Bernard, T. (2022). "*Analyse haute résolution de la morphologie des paysages et des processus à partir de LiDAR aéroporté répété et simulation hydraulique"* (Doctoral dissertation, University of Rennes).

INTERNATIONAL CONFERENCES

- **July 2025 (Lyon, France) :** Bernard, T., Pandrin, E., Bertoldi, W., Capito, L. & Bizzi, S.: " *Bedload and morphological active widths of gravel-bed rivers: variability among timescales*". In *EGU General Assembly Conference Abstracts* (pp. EGU25-4044).

- **April 2025 (Vienna, Austria) :** Bernard, T., Pandrin, E., Bertoldi, W., Capito, L. & Bizzi, S.: "*Variability in bedload and morphological active widths of gravel-bed rivers across timescales*". In *EGU General Assembly Conference Abstracts* (pp. EGU25-4044).

- **April 2021 (Vienna, Austria):** Bernard, T., Davy, P., & Lague, D. : "*A hydrodynamically consistent" slope-area" relationship for analysing fluvial landscape with wide rivers*". In *EGU General Assembly Conference Abstracts* (pp. EGU21-8423).

- **December 2019 (San Francisco, United-States):** Bernard, T., Davy, P., & Lague, D. : "*How 2D Hydraulic Modelling Can Improve Landscape Analysis in Tectonic Geomorphology?*". In *AGU Fall Meeting Abstracts* (Vol. 2019, pp. EP31C-2301).

- **April 2019 (Vienna, Autria):** Bernard, T., Lague, D., & Steer, P. (2019, January). "*Detection and volume estimation of earthquake-triggered landslides from multitemporal LiDAR and tri-stereo satellite 3D remote sensing*". In *Geophysical Research Abstracts* (Vol. 21).